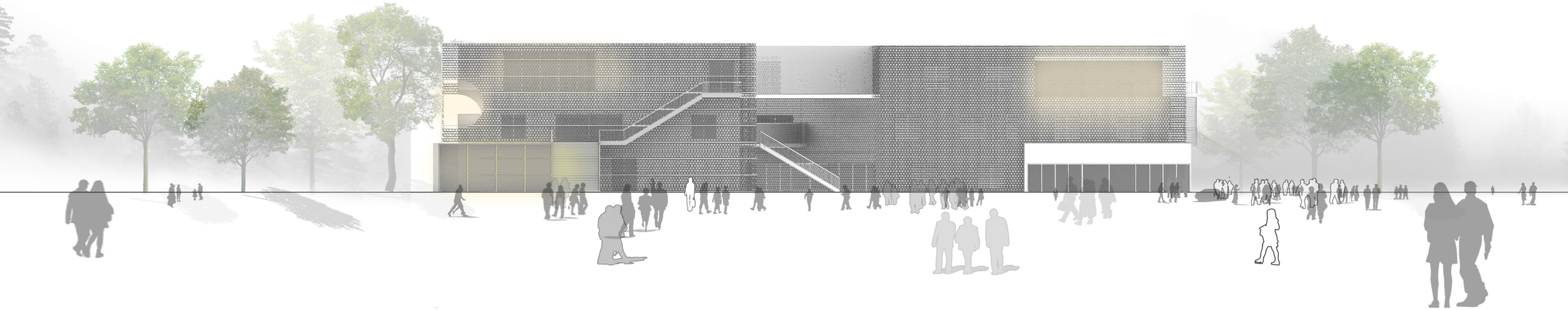


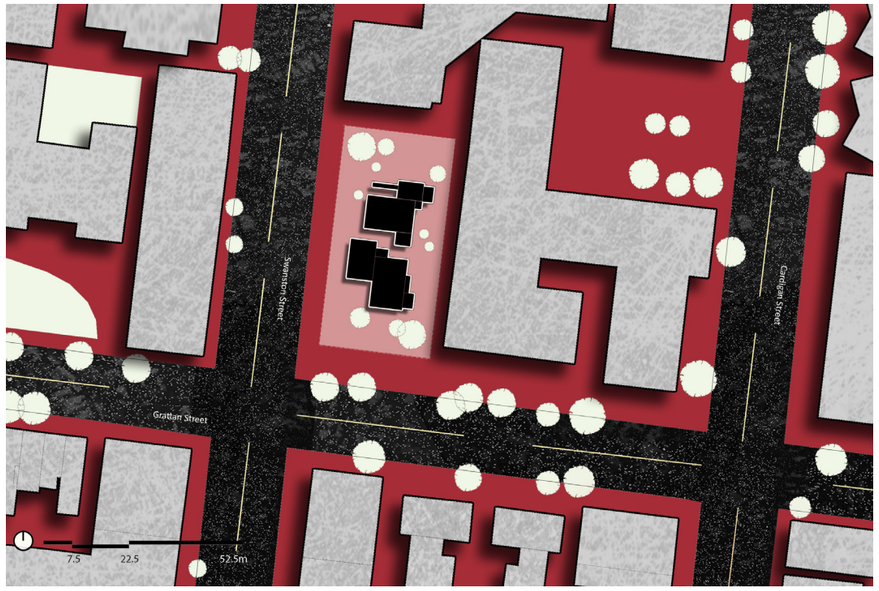
ARCHITECTURE PORTFOLIO

Undergraduate

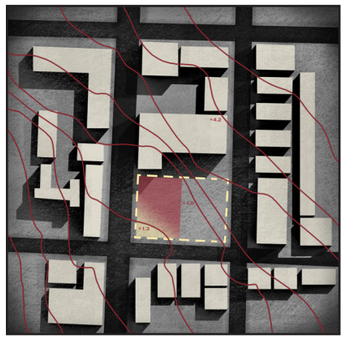
Shiran Geng

1.STUDIO C

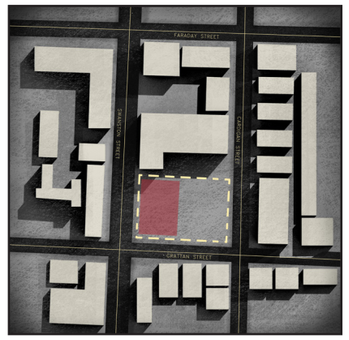
700 SWANSTON STREET – GALLERY FOR THE BROKEN HEARTS



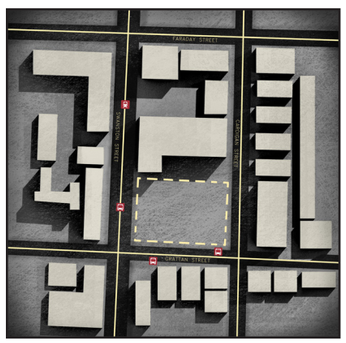
SITE PLAN



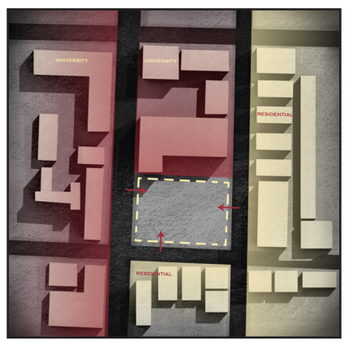
TOPOGRAPHY



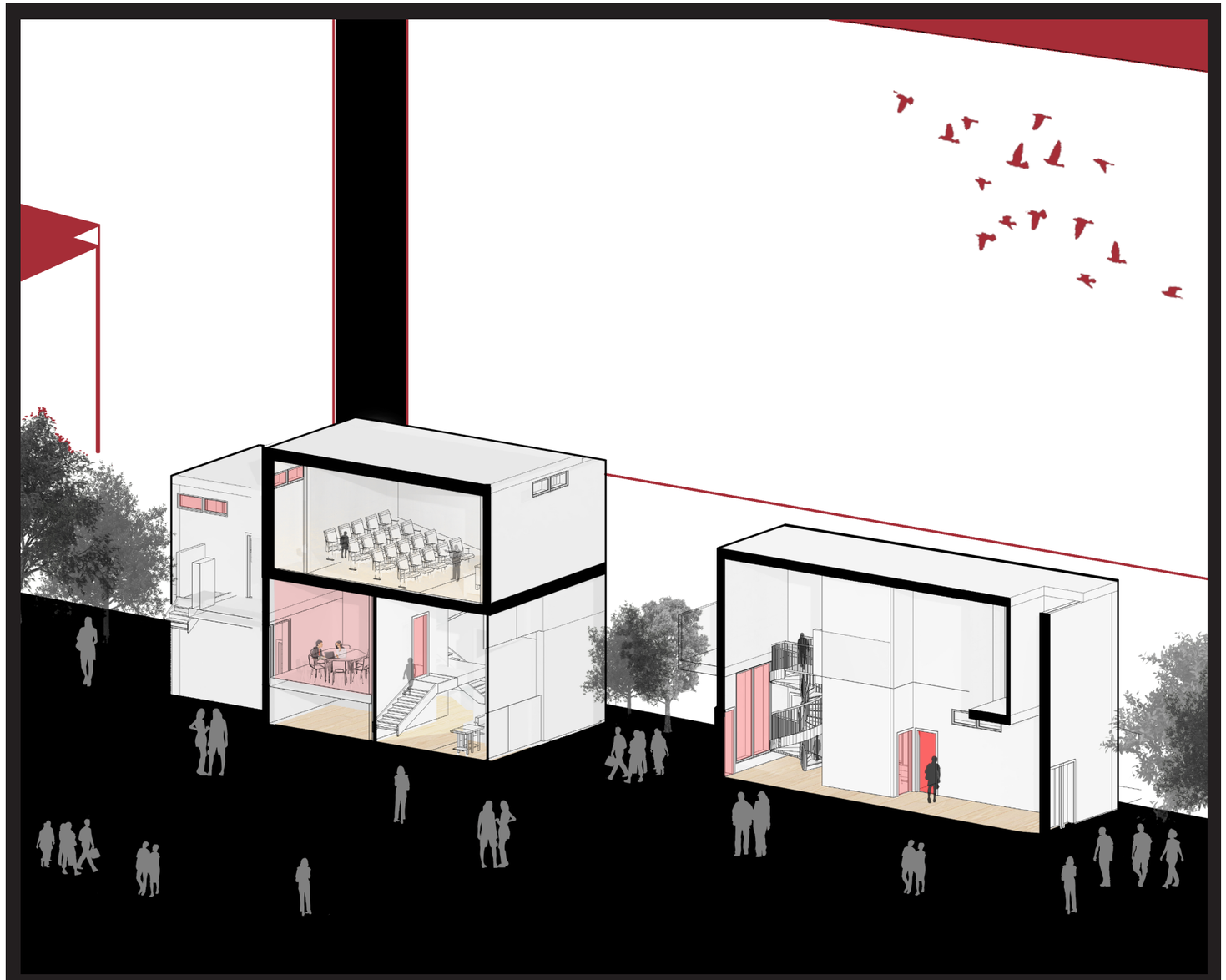
SITE BOUNDARY



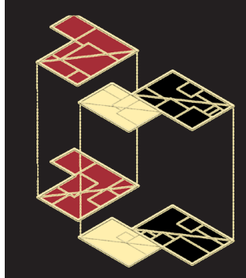
ACCESS



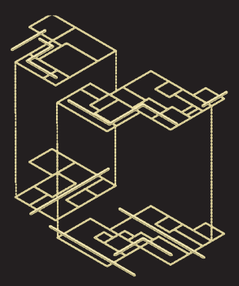
USERS



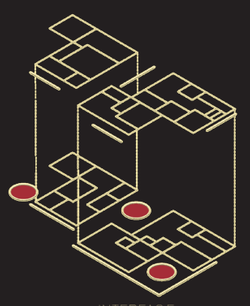
SECTION AA PERSPECTIVE



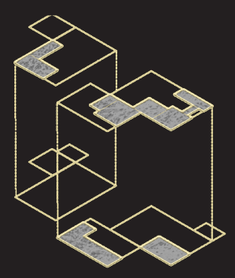
UNIT



MOVEMENT

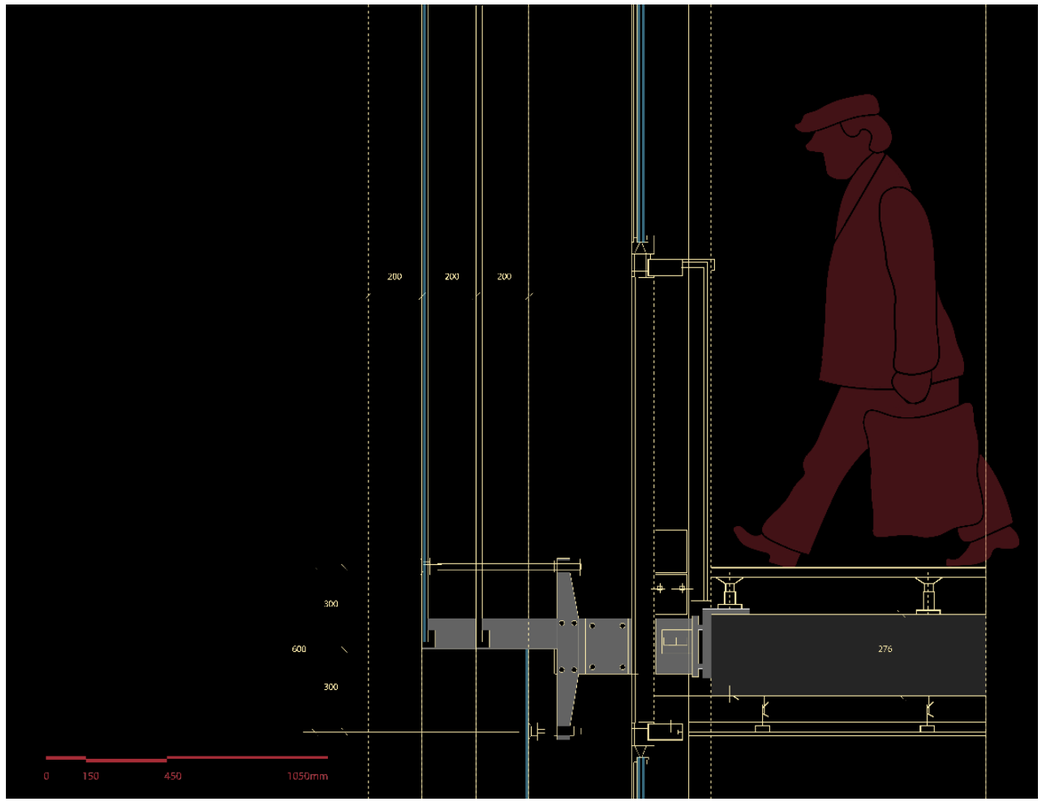


INTERFACE

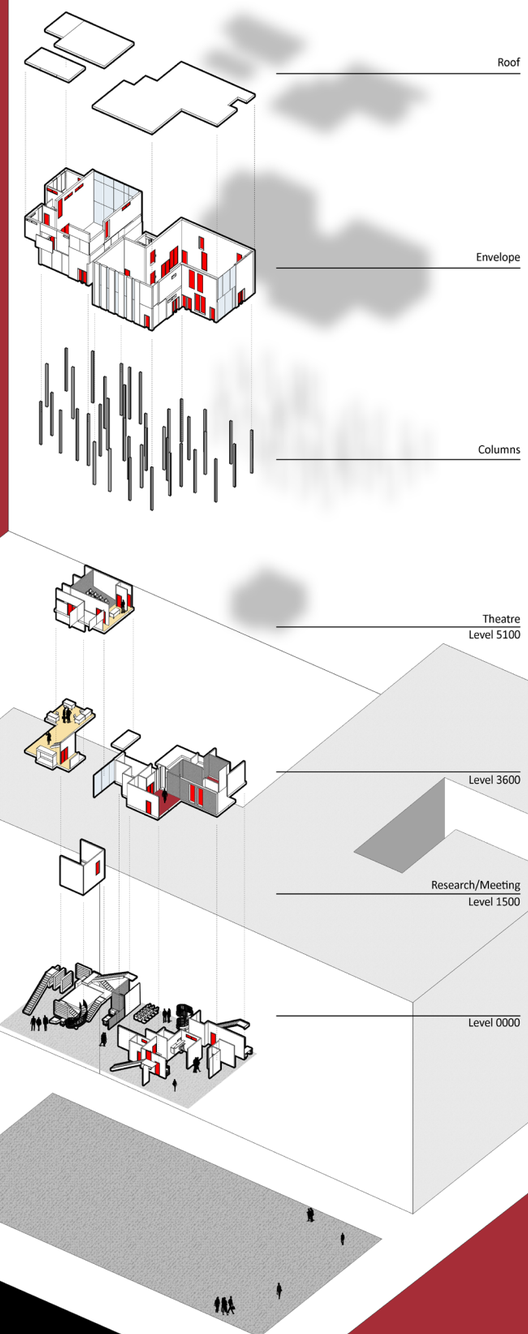


GALLERY

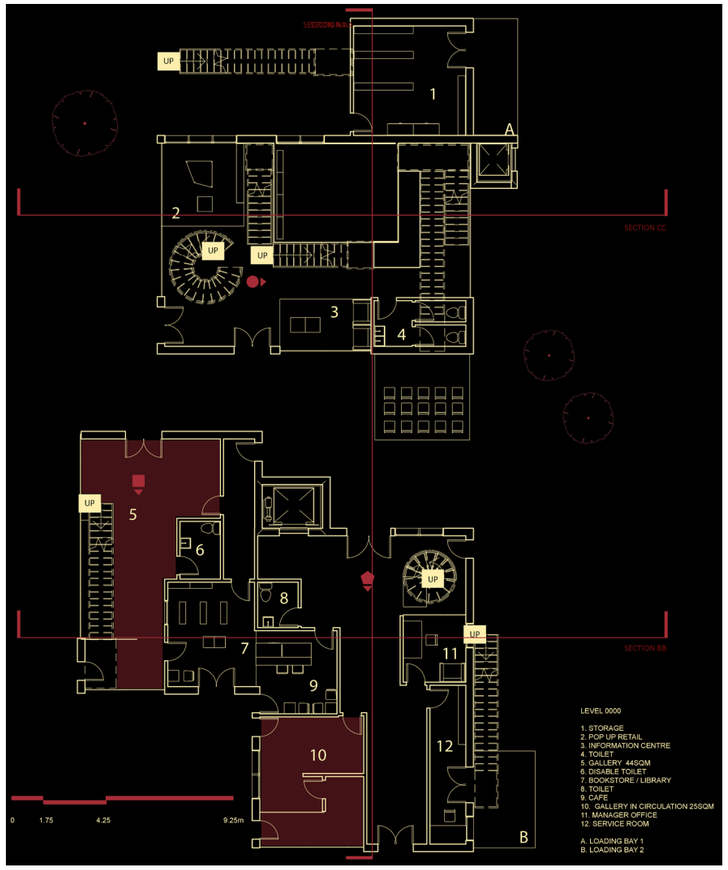
SECTION AA

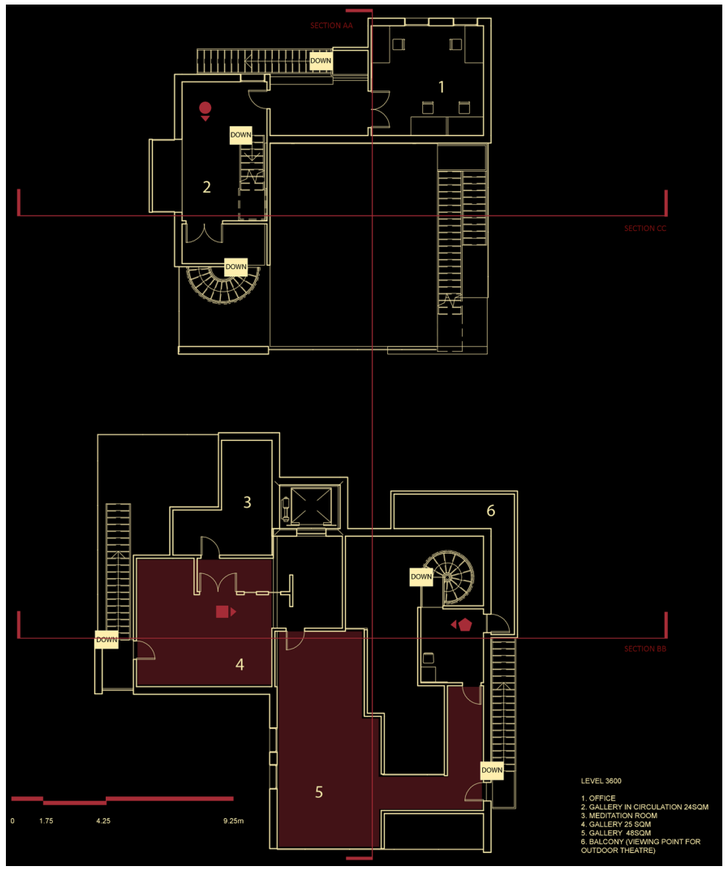


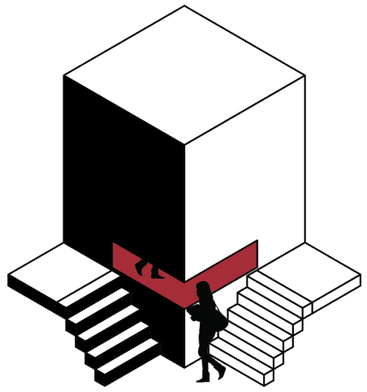
PORTAL SECTION



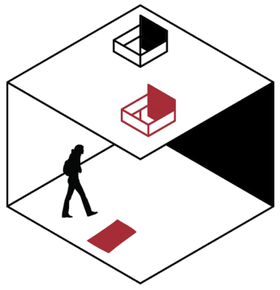
ISOMETRIC



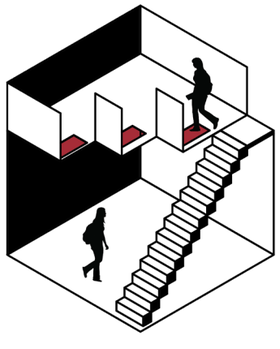




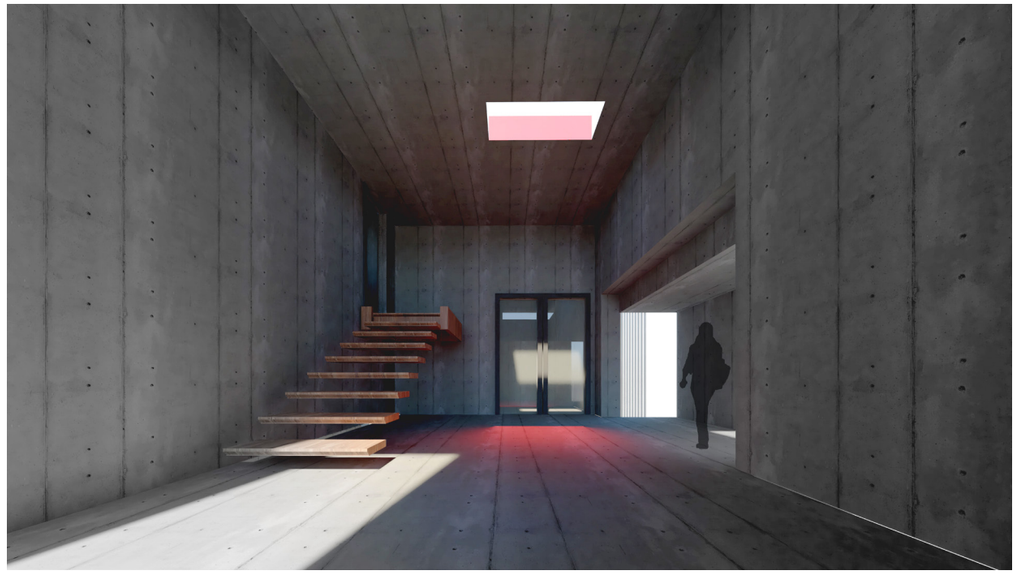
Past:Ground

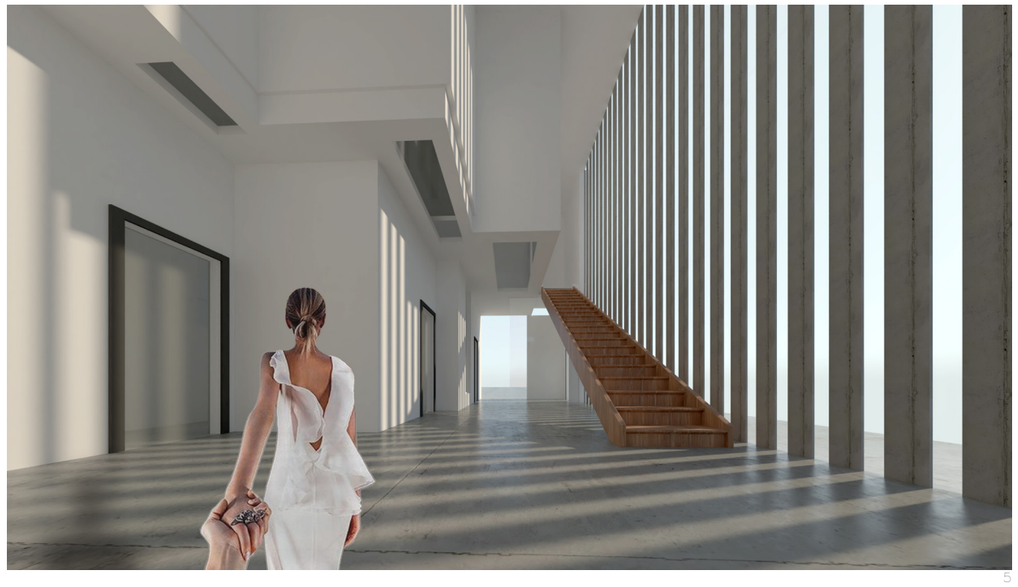


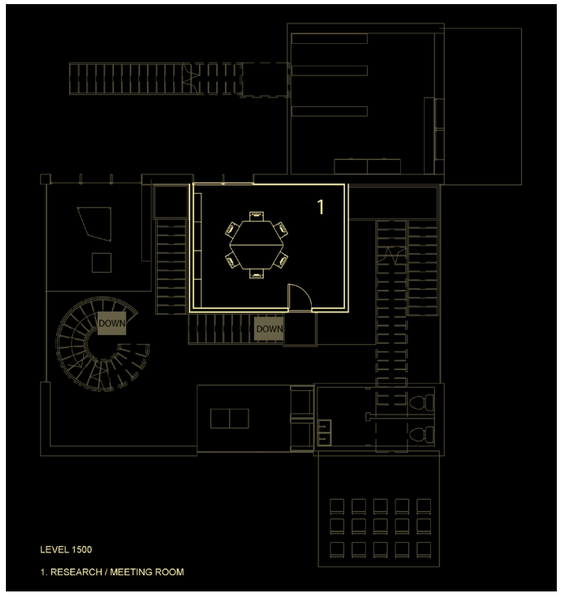
past:first

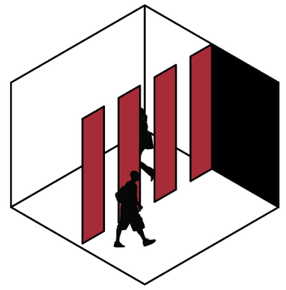


present:ground

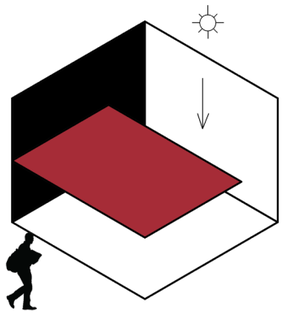




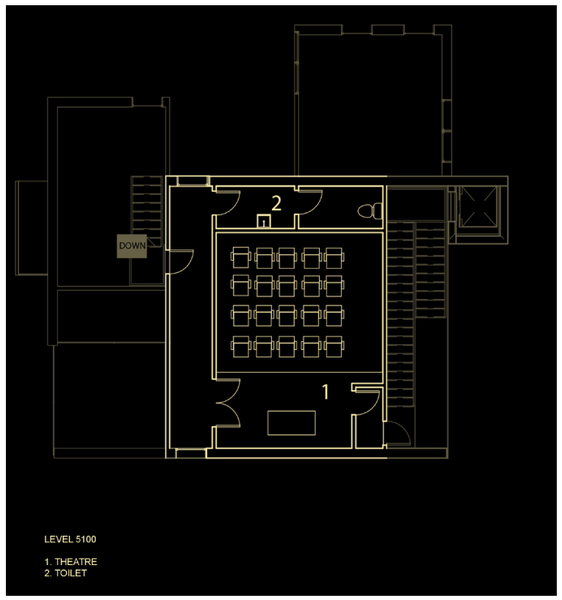


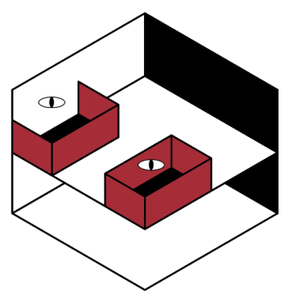


present:first

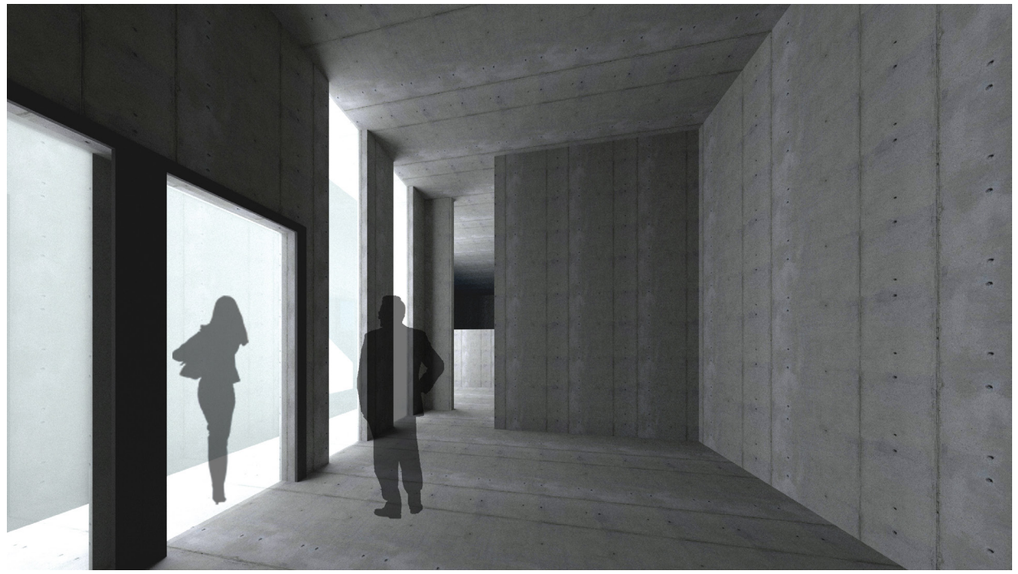


future:ground





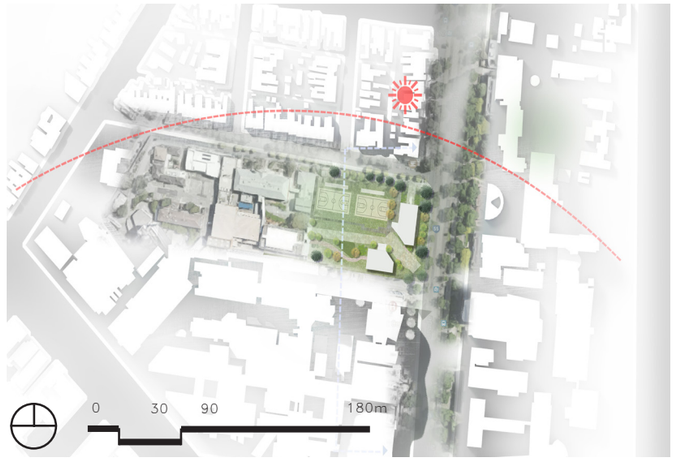
future:first

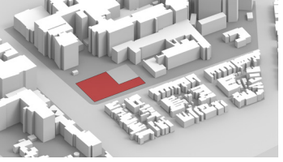


1.STUDIO FIRE

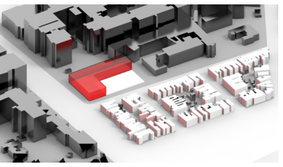
University High School – New School

The project brief is to design a new learning Centre complex as part of the University High School, Story Street, Parkville. The existing facility of school does not include much community space and there is a lack of interaction with the public. The project concept is drawn from shifting viewports, expanse of mist and water from ancient Chinese paintings. The building is designed as a gallery of learning, where the activities of the users are put on display for all to see, aiming to encouraging the interactions and stimulating users interests in exploring knowledge. Two major units are designed to serve for different functional purposes, representing the concept of different perspectives. The south unit is mainly for school students and staff whereas the north unit is for community users. The programs are arranged in a manner that users can constantly see others activities and have a different perspective as they move across different programs. The mist and water are expressed on the building façade, where the steel panels acting as not only the shading device but also creating a sense of ambiguity, mystery and curiosity for users.

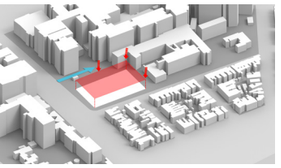




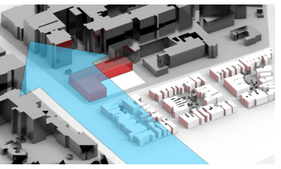
SITE BOUNDARY



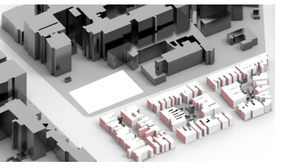
POTENTIAL SITE PARTI



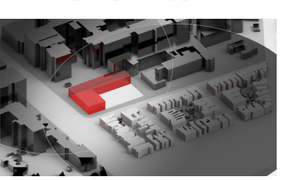
VERNICULAR ACCESS



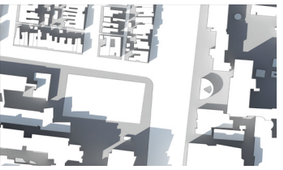
NATURAL ENVIRONMENT



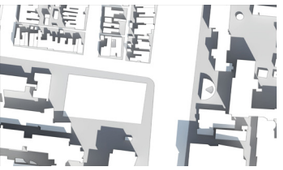
MATERIALITY IDENTITY



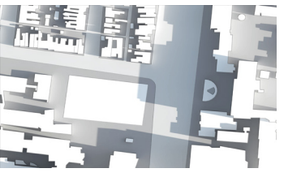
COMMUNITY INTEGRATION



SHADOW STUDY LATE AFTERNOON

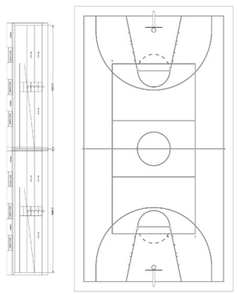


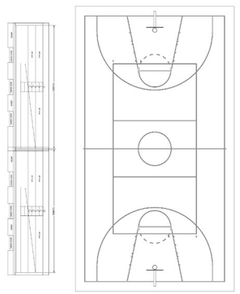
SHADOW STUDY MORNING–JUNE



SHADOW STUDY AFTERNOON–JUNE

PROGRAM ANALYSIS

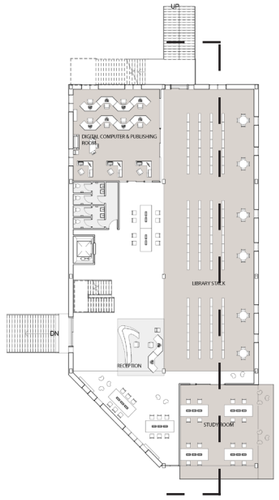




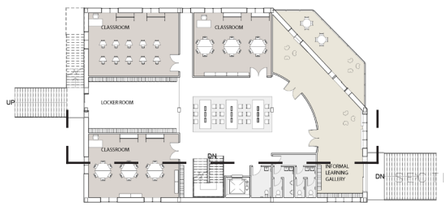
SECTION A

SECTION B

GROUND LEVEL



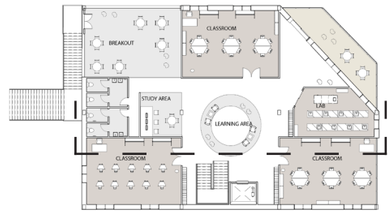
SECTION A



SECTION B

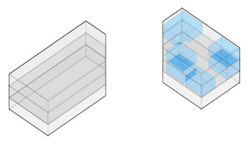
LEVEL ONE

SECTION A

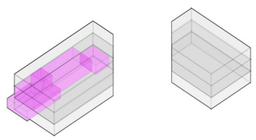


SECTION B

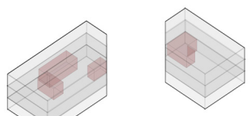
LEVEL TWO



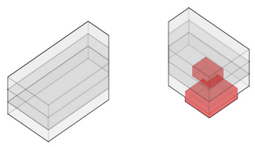
TEACHING SPACE



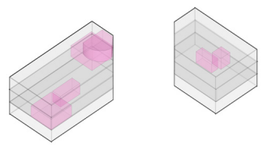
SPORTS AREA



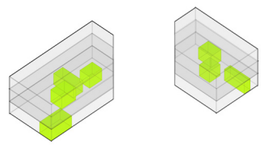
INFORMAL LEARNING



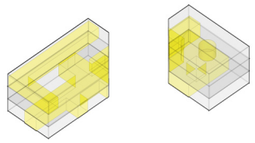
BREAK OUT AREA



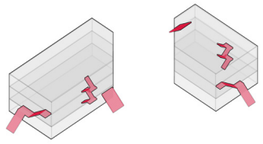
STUDY SPACE



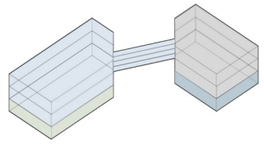
STAFF OFFICE



INFORMAL LEARNING GALLERY SPACE



CIRCULATION

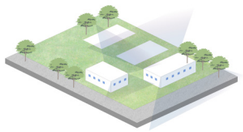


MAIN FUNCTION ZONES

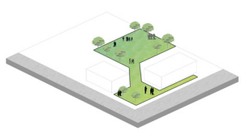
SECTION A

SECTION B

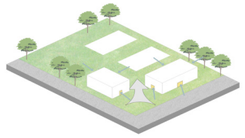
SITE INTERACTION DIAGRAM



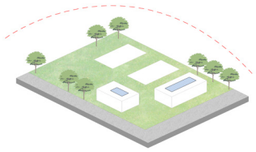
VIEW



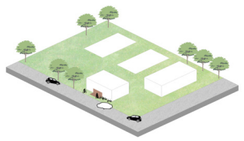
MAIN GREEN ACTIVITY AREA



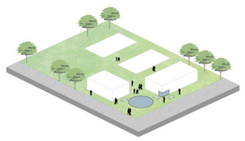
SITE ACCESS



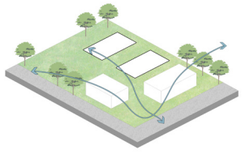
LIGHTING&SKYLIGHT



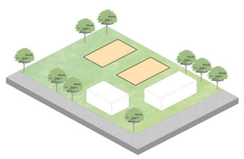
PARKING



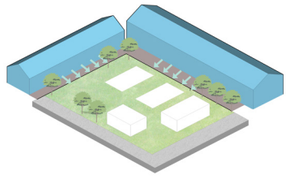
SITE INTEGRATION
(ROYAL PARADE)



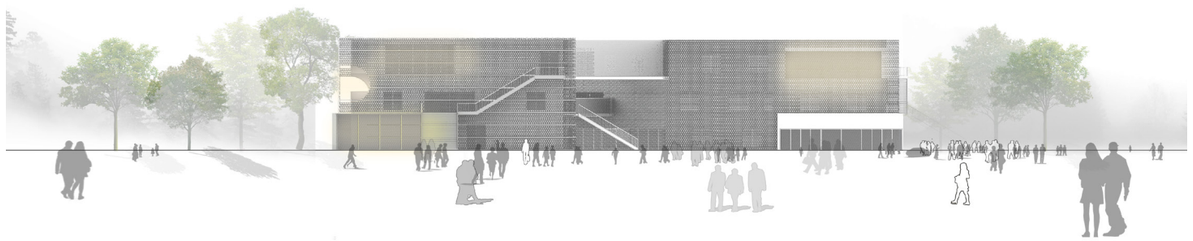
MAIN SITE FLOW



SPORTS AREA



SITE INTEGRATION
(RESIDENTIAL)



WEST ELEVATION

CLASSROOM

GYM

EXPLODED ISOMETRIC

LEVEL TWO

Building A:classroom, lab, breakout

Building B:gallery, cinema, meeting rooms

LEVEL ONE

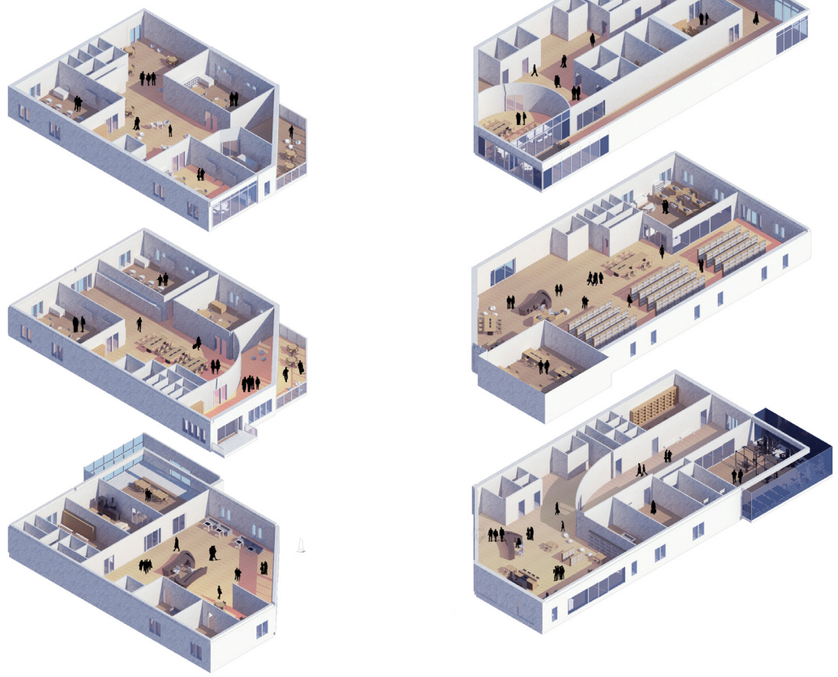
Building A:study space, classroom,informal learning space

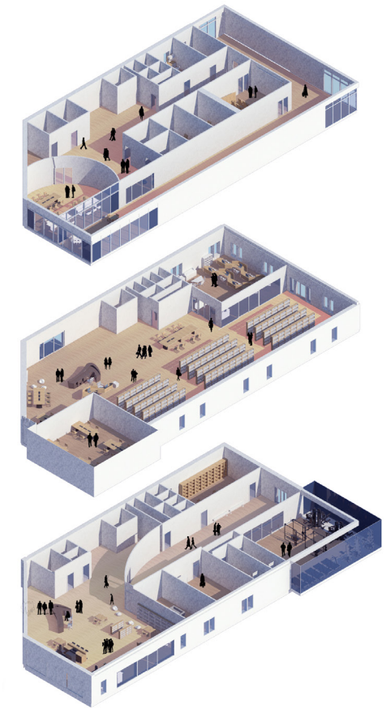
Building B:library, publishing room, study room

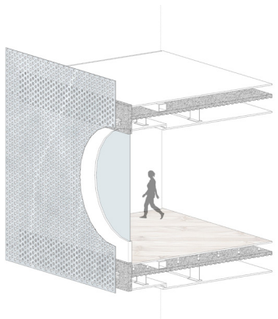
GROUND LEVEL

Building A:reception, breakout, staff room

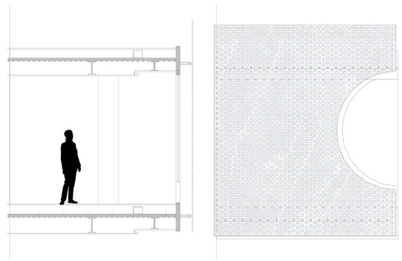
Building B:atrium, office, gym







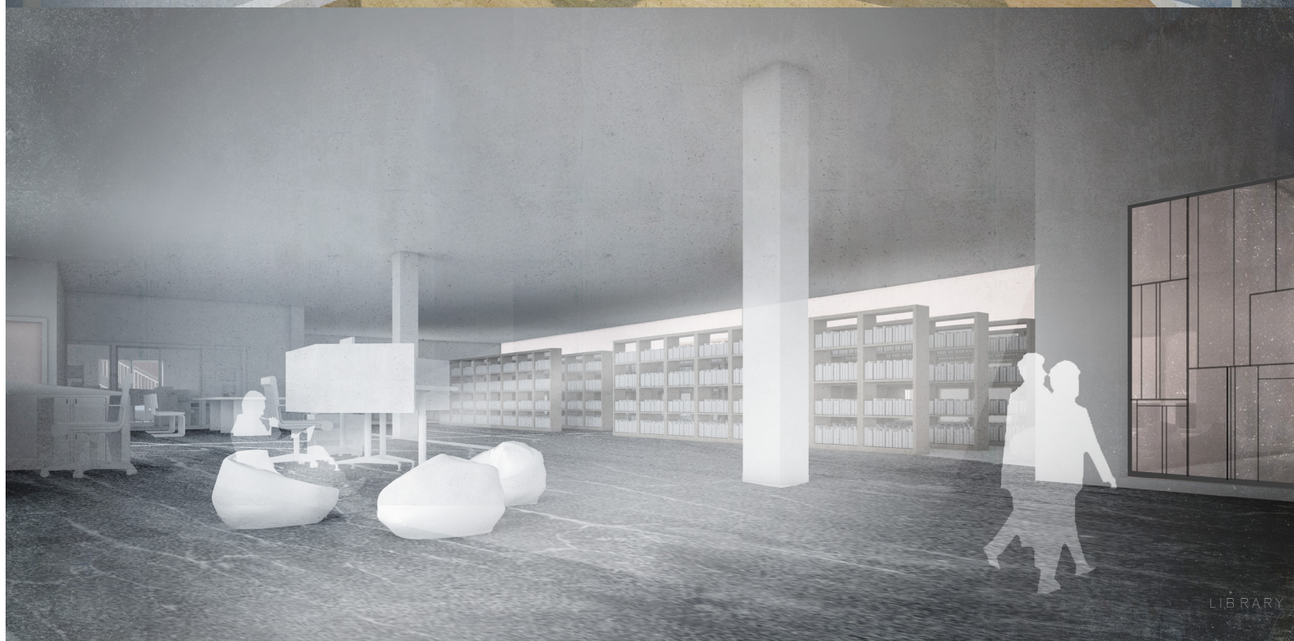
SECTION DETAIL PERSPECTIVE



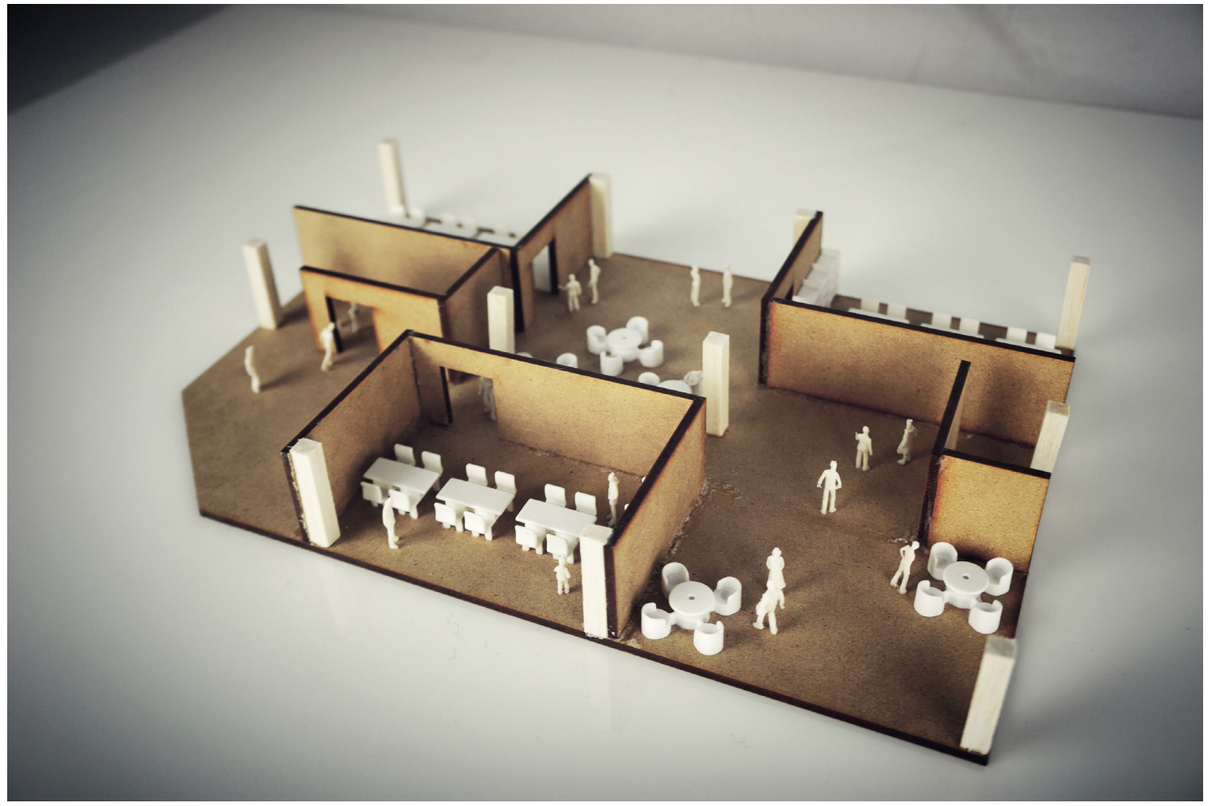
SECTION DETAIL

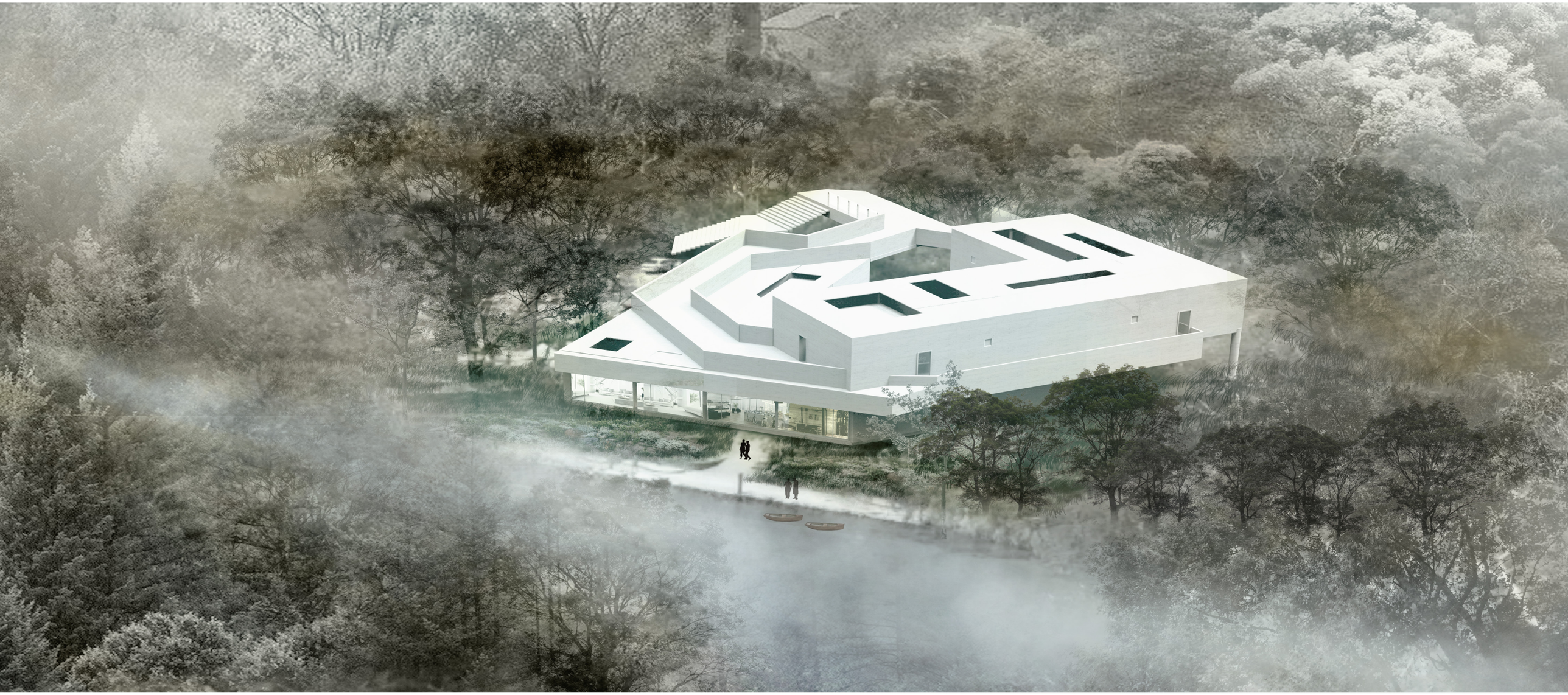
GALLERY

BREAKOUT SPACE



LIBRARY

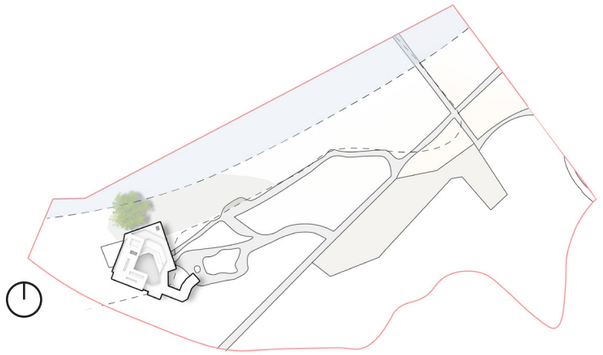




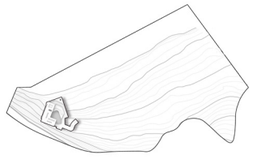
2.STUDIO WATER

STUDLEY PARK BOAT HOUSE

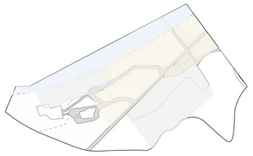
This project is located at Studley Park in Kew. The design proposes a replacement for the existing boat house which needs to be upgraded to current needs of the site users. My concept is to create a boat house where users recreate as well satisfying the primary functions that the brief required. My design functions around a few moments inside the building, the boat storage which acts as a gallery space, the open kitchen restaurant and the café that has a private view on top of the building. The level of privacy increases with the floor level – from complete public on the ground level to complete private on the third level where the little café is located. By separating the building with privacy level and permeability, users can experience different level of relaxation with their choice. By creating a lot of openings and free circulation that has a function comes with it, the building has a flexibility feature which create some key moments of relaxation for the users.



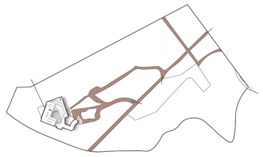
Site Plan



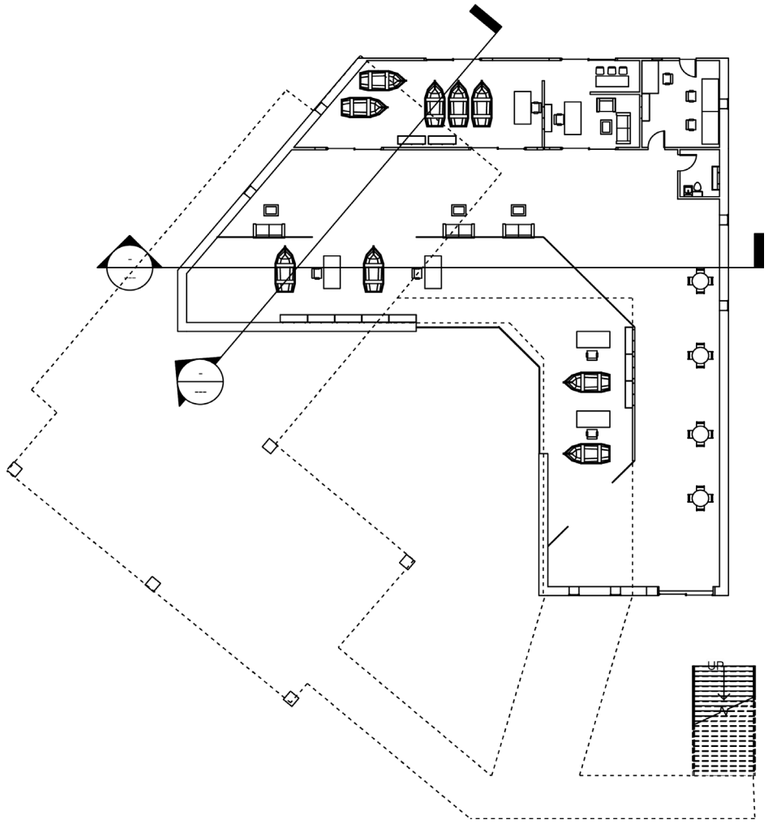
Topography



Function Zoning

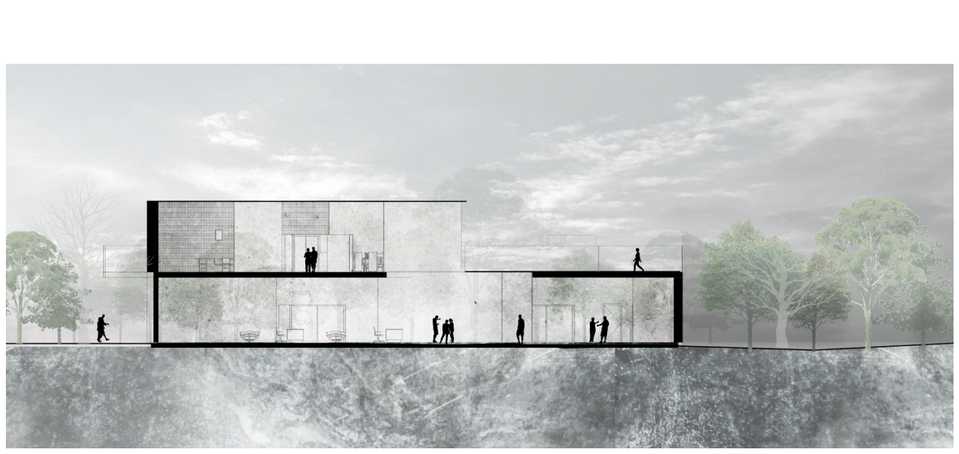


Site Circulation

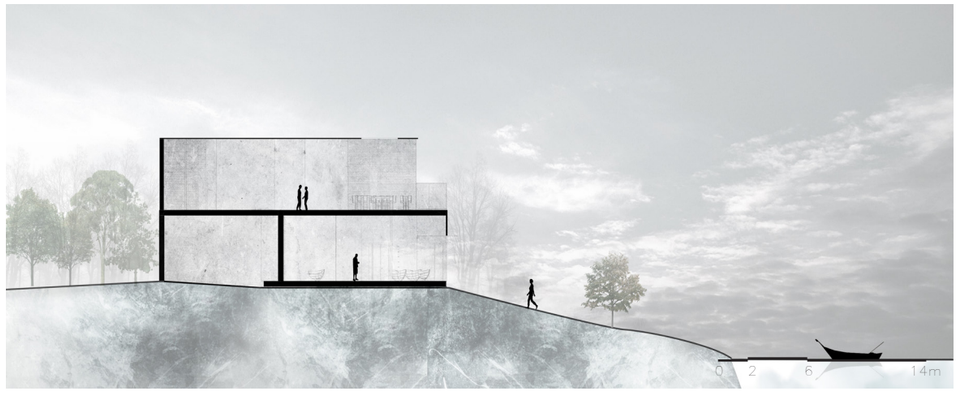


Ground Level

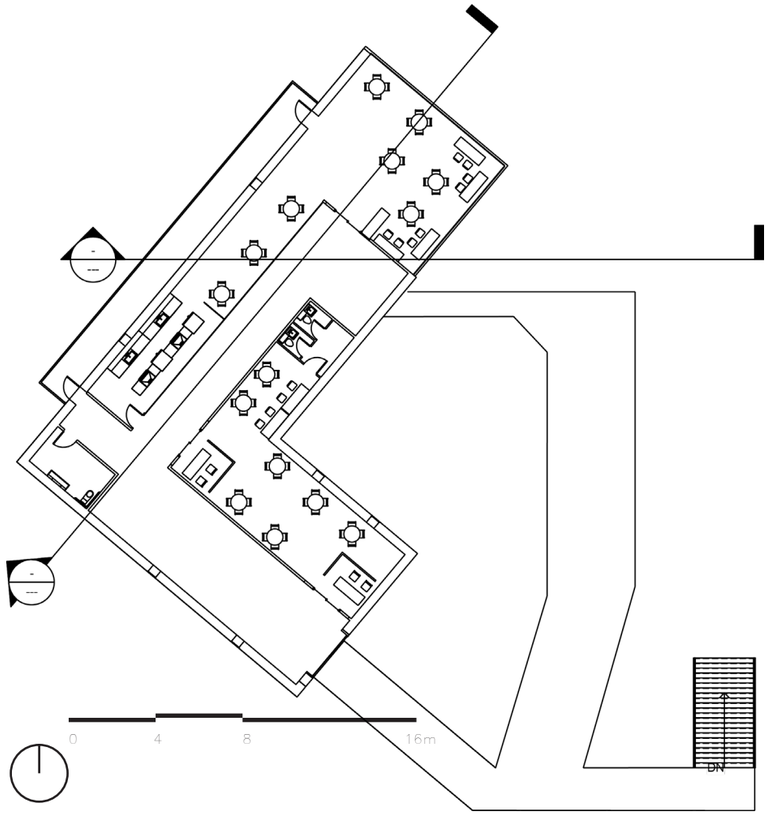
1. Boat storage
2. Boat hire
3. Office
4. Workshop
5. bathroom



SECTION 1

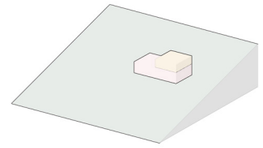


SECTION 2

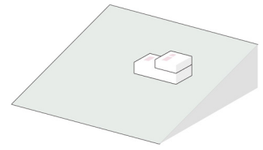


Level One

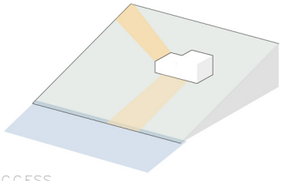
1. Cafe
2. Kitchen
3. Resturant
4. Bathroom



FUNCTION AND LEVEL
ground level includes functions that are more public whereas level one contains fucntions that can be used for more private purpose

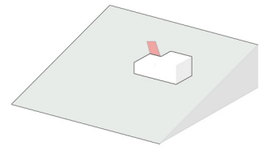


CEILINGS
Both ground level and level one ceilings have mutiple skyligts for the natural light to come through

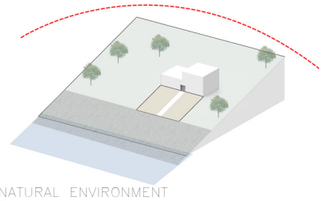


ACCESS
Two major pathways are provided to the building. One is for visitors from the site, another one is for boats from water.

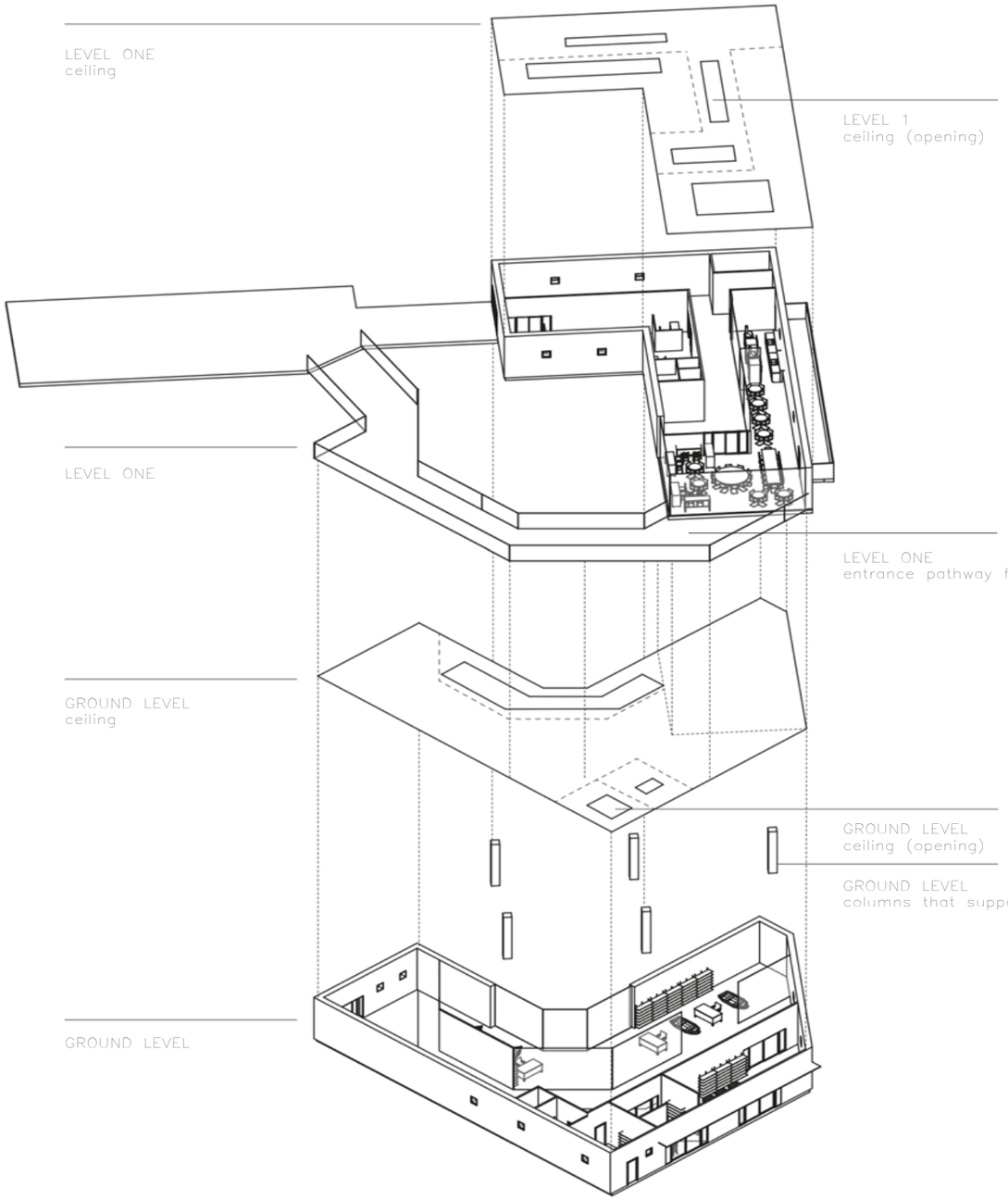
VIEW
Two water facing openings are created to preserve the view for visitors. These two major openings also provide the building a better ventilation.



VERTICAL CIRCULATION
Due to the topography of the site, outdoor stairs are needed for people to access level1.



NATURAL ENVIRONMENT
Overall, this boat house works well with the natural environment on site. It preserves the water view and getting the most sunlight through mutiple skylights.



Exploded Isometric

Interior 1

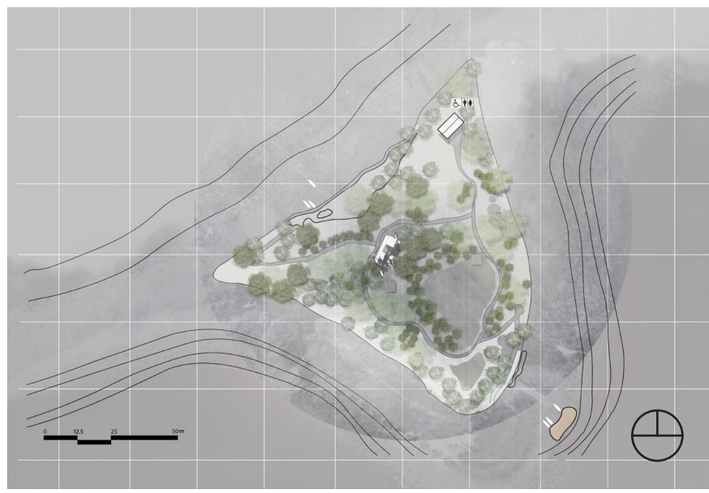
Interior 2

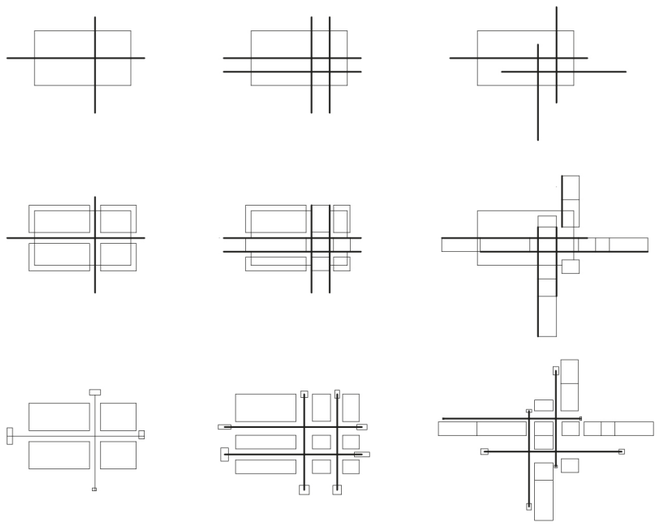
Exterior

4.STUDIO EARTH

REALISING AND HIDING A SECRECT

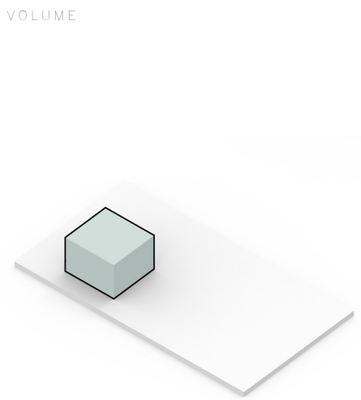
The brief focus on designing a place to store a secret. The site is Herring island, which is an island in Melbourne offers recreational and cultural opportunities. My design is about how the space could have a sense of navigation that lead the users to reveal the secret through the journey that they create for themselves. In the end, the secret that the users think is hidden behind the walls is actually all in their mind and is often the best way to keep secret. In order to create the secret revealing journey, I started to look at how to generate a sense of curiosity for people in order to lead the people in their spatial exploration without any direct signs of navigation. I then created walls and different atrraction points that helps to guide visitors in their journey of realizing and hiding the secret.



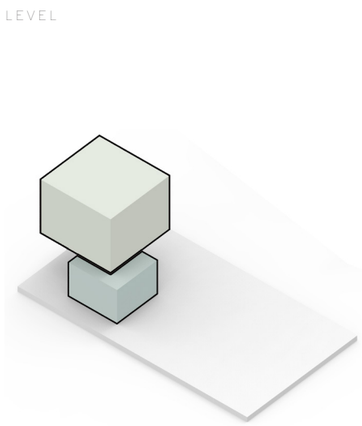


FORM FINDING: UNITY/SEPERATION/CONNCETION

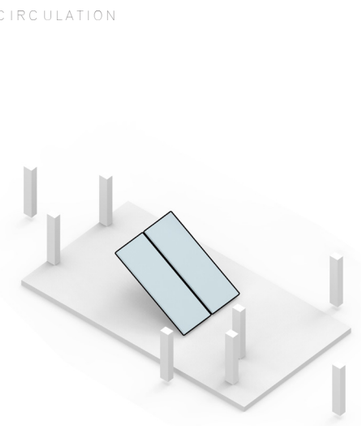
The relationship between point, line and plane are shown on a set of diagrams. First, I cut through the complete forms with lines. Then, I separated these forms with some distance. By doing this, I understand more of the separated forms. Finally, I connected these lines with finishing points. After this process, I then decided how to place these forms and connect them with lines and continue using the point, line and plane concept throughout my design.



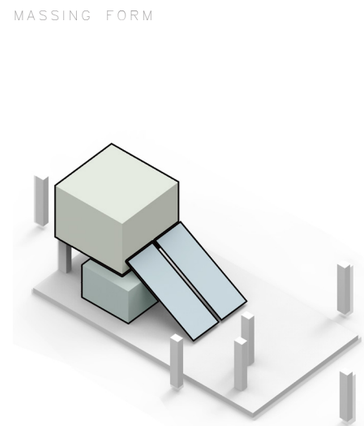
The building form is based on one cubic shape on ground level



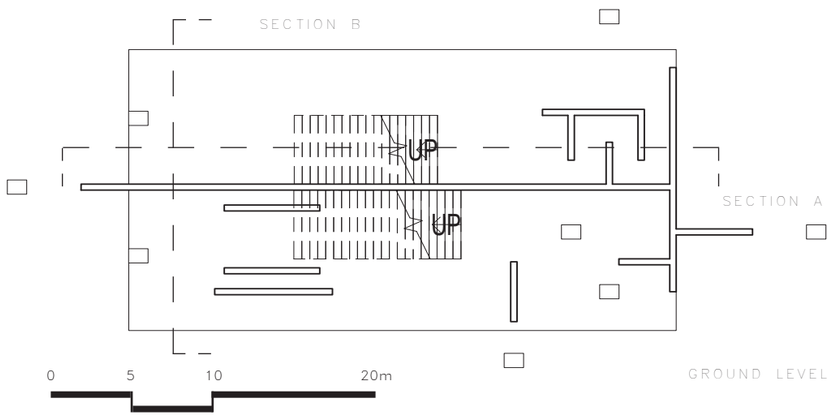
On level one, another cube is added to the form and together forming two levels

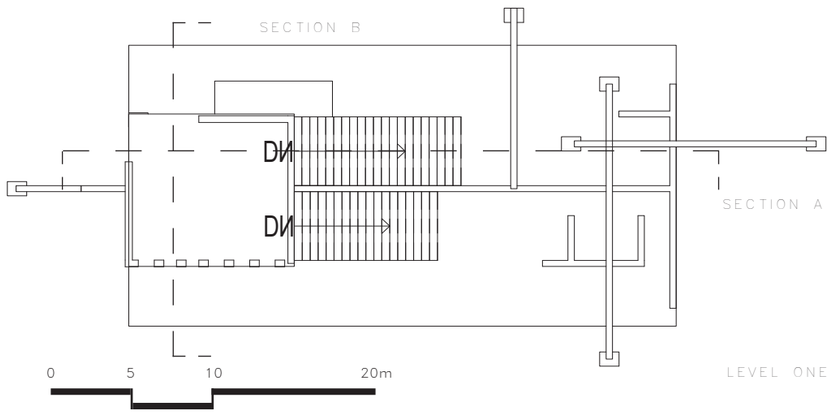


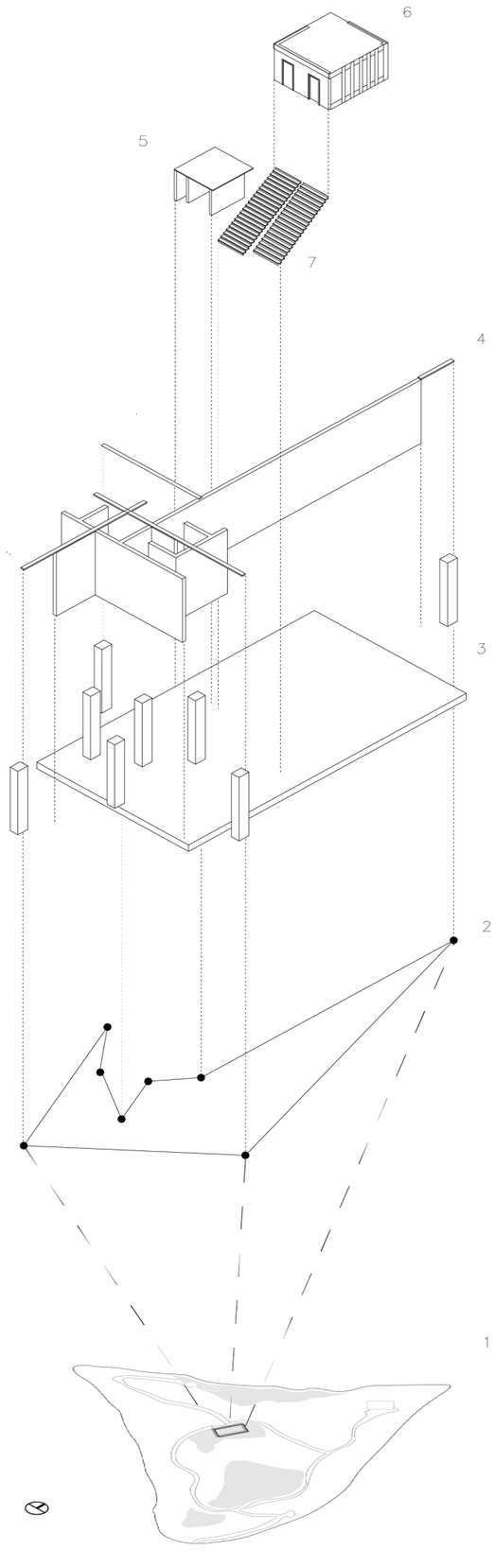
Stairs are used to connect two levels, forming the journey



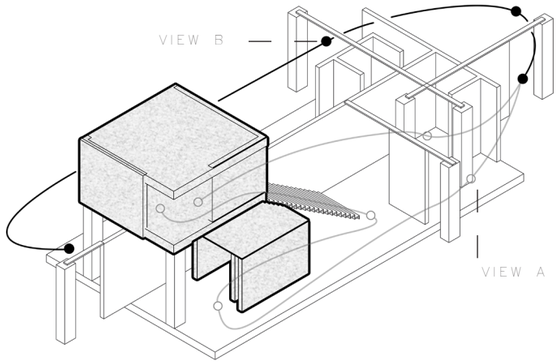
Massing study outcome





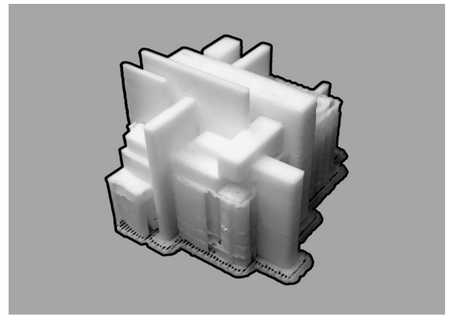


1.site map
2.points generated from form making
3.columns according to points posistion
4.line and plane form points connected wall panels
5.secrect unit I
6.secrect unit II
7.pathway from secrect unit I to secrect unit II

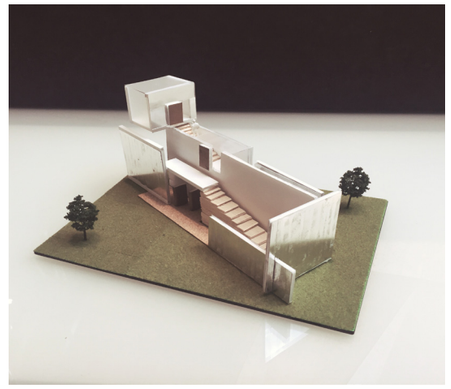


SECRET JOURNEY

This diagram shows a dotted map of each attraction point in my design. At each attraction point, user will feel the desire of being behind the wall to find out what is the hidden scene. A journey full of secrets is then built and connected by these attraction spots



CONCEPT 3D PRINT

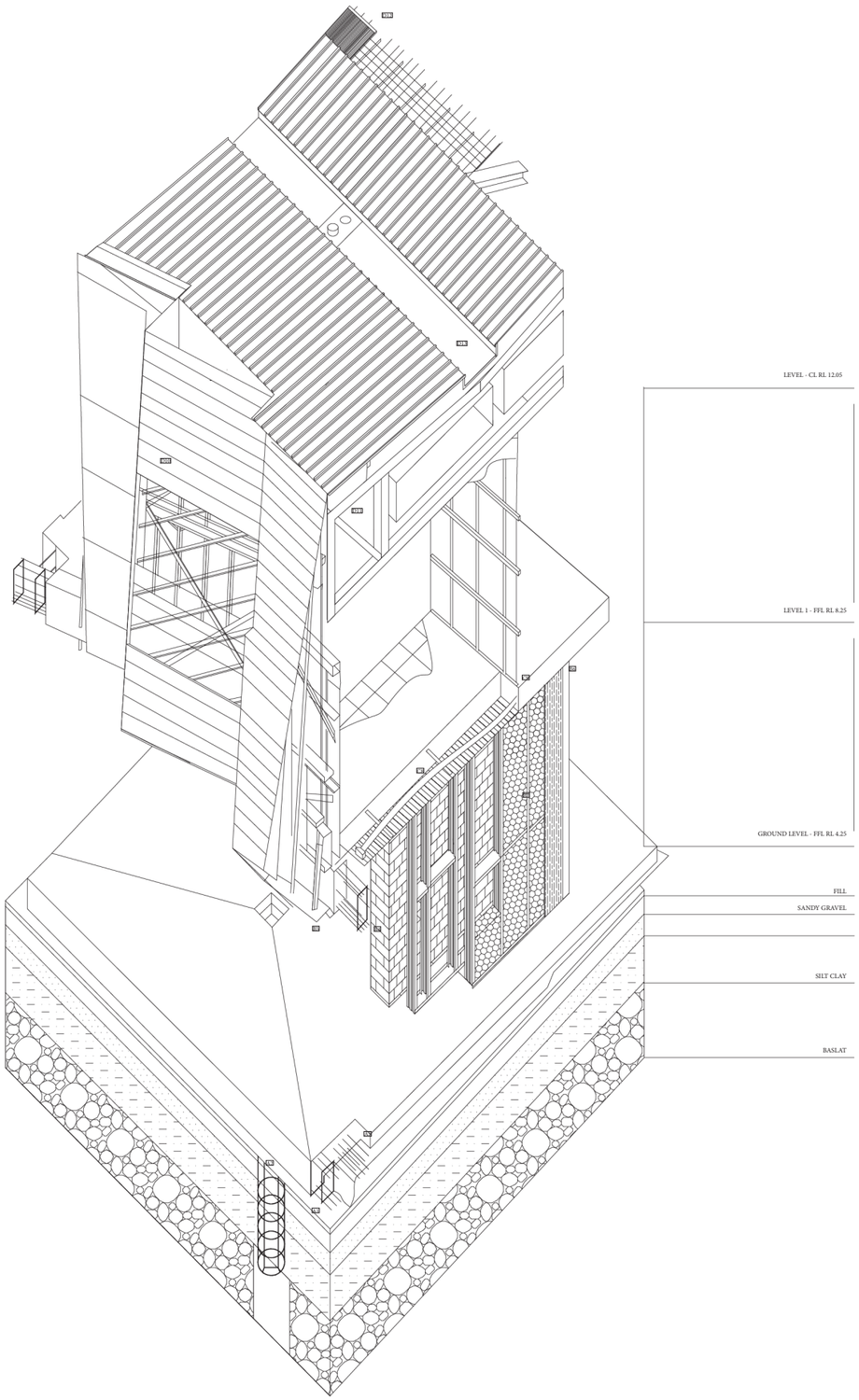


CONCPETUAL MODEL

VIEW A DURING THE JOURNEY

VIEW B DURING THE JOURNEY

CONSTRUCTION DESIGN



A1.Ground beams
13n16 are the reinforcement bars type.
Blinding concrete is used where fill are not level.

A2.Bored pile
The bored pile extends downwards until the weathered basalt rock.
The bored pile extends at least 12mm into the weathered basalt. Bored concrete has 40mpa strength and a diameter of 400

A3.Floor slab
Floor slab is 150 mm thick. Reinforcement bar are placed along the bottom and the top of the floor. These reinforcement bars are held by bar chairs as the concrete is being poured. A 0.2mm thick flexible polymetric film is placed under the slab as a waterproofing membrane.

B4.Level 0 external walls
The external walls are brick veneer.
Anchor ties are used to connect the brick to the stud framing as it is part of the envelope of the building.
A waterproofing membrane lines the steel stud frame.
A cavity flashing is used.
Austral bricks are used, with a zinc' with a graffiti coat applied to the exterior.

B5.Level 0 external ceiling
The external ceiling at level 0 is a ply substrate. It is connected to the structure by 22mm top hats along the base of the slab. A external breathing membrane and a aluminum sill at the edge.

B6.Internal ceilings
The ceiling is a suspended plasterboard bulkhead and lined with insulation.
The ceiling is held up my clips, rods, furring channels and cross rails connecting to the framework of the level 1 floor.

B7.Internal walls
Walls in this section consist of steel stud framing.
Internal walls consist of batt insulation within stud frame.
A plasterboard lining before the finish is applied.

C8.Level 1 beam
Concrete beams run through the middle of the level one floor and along the edge.
Regular reinforcement, post tension reinforcement are used with in the concrete beam. Because some parts of the concrete are acting primarily in tension and the dead load of itself may cause failing.

C9.Level 1 floor slab
A bondek is placed under the concrete slabs of level 1 flooring.
This means less concrete is used, making it more economical, less spacing consuming and lighter.
A steel stud frame is placed under the bondek in order to hold it up. Concrete screed is used as floor finishing.

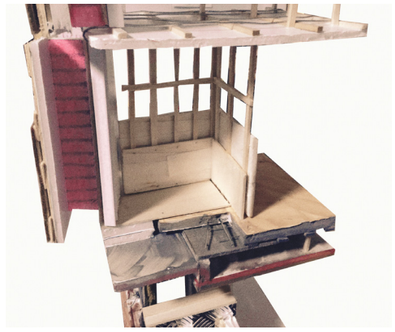
C10.External wall
Columns are hot rolled steel I beam are the main structure of the walls
Zigzag shape is made in the wall with different claddings as a non-structural member (decorative).
The mc4 wall consists of zinc metal cladding with a aluminum rain screen
Waterproof membrane and furring channels are placed behind.
Permanent steel cross bracing is needed because the diagonal angles placement of the I beam. Steel framing and insulation are placed inside.
For the internal side of the wall, plaster board is used as a finishing.
The mc1 wall has a rain screen and a weather proof membrane.
Top hats attach to the cladding and wall girts attach the cladding to the steel stud frame. Insulation sits inside the steel stud frame and a plasterboard covers the frame and then covered by the finishing.

D11.Roof structure
The roof structure consists of various roof trusses with diagonal webbing and a hot rolled steel frame.
Full strength butt welds, cleat plate with continuous fillet welds and web stiffeners are the main connecting methods.
Ventilation as a roof system service runs throughout the structure.

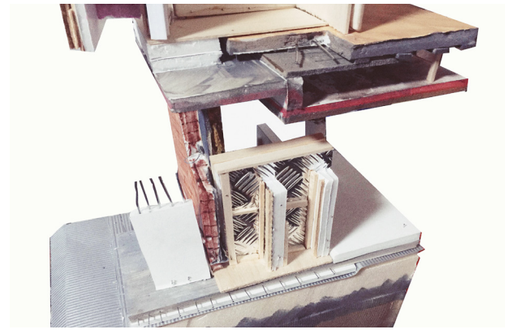
D12.Roofing
lysaght kliplock zincalume roof sheet is the roof sheet type.
Below the sheet, there are a breathing membrane and a vapour permeable membrane.
Connected to a plywood backing that is placed on top of the 75mm fletchers roof rack system.
All these systems listed above are placed on top of the z purlins that span across the roof.
Between these roof rack systems, blanket insulation is used. A reflective foil underside and a safety mesh are placed below.

D13.Box gutter
Bluescope lysaght is the box gutter type. It is made out of zincalume.
A spandeck support that span across the gutter, supported by an angle frame suspended from the roof purlins.
Flashing over the ply board is necessary. The box gutter has its own insulation

Sriran Geng 666071	Construction Design
Glasshouse Construction Drawing	Assignment One
	Section J



LEVEL ONE DETAIL



GROUND LEVEL DETAIL

FINAL MODEL